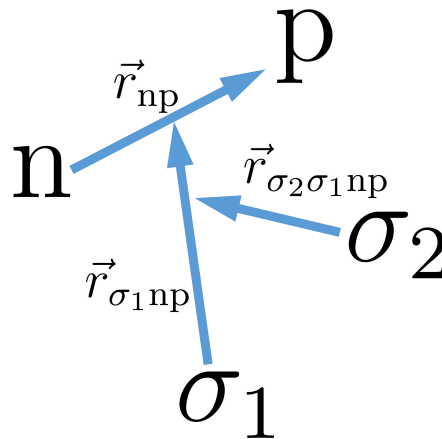


Effects of the Second Meson in Deuteron Model with σ -Mesons

(Effekten af den anden meson i deuteron-model med σ -mesoner)

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1 Abstract

Nuclear forces are of particular interest, since in eight decades of research, it is still not possible to model nuclear structures on the basis of a basic interaction. In practice phenomenological effective potentials are used to model nuclei, but in [1] a nuclear model is proposed, where the nucleus is in a superposition with mesons, that are treated explicitly.

In this thesis the model is used on the simplest nucleus, the deuteron, dressed with up to 2 scalar-isoscalar σ -mesons, yielding a four-body problem to solve. This is carried out numerically, using the correlated Gaussian method. Computations in [1] already suggest a bound state for the deuteron with one σ -meson. As expected, it is found that the second meson gives an additional binding term to the deuteron ground state energy. For reasonable choice of parameters, such as a meson mass of $m_\sigma = 500$ MeV, that yield the experimental binding energy of -2.2 MeV, the second σ -meson contributes with -0.173 MeV or about 7% of the total binding energy. It is shown that the second meson's contribution increased, by decreasing the meson mass m_σ down to masses lower than pion masses, at about 137 MeV.

The second meson gave just a small correction to the deuteron's wave function found by the correlated Gaussian method. With this method the wave function automatically became symmetric to the exchange of the mesons.

The outlook of this research is that a third σ -meson only would contribute with about or less than 1% to the total binding energy of the deuteron. Furthermore the increasing significance of the second meson for lower masses, suggests that a nuclear model including at least two pions should be investigated.

2 Resumé på dansk

Kernekraften er specielt interessant at arbejde med, da det efter firs års forskning stadigvæk ikke er muligt, at modellere atomkerner vha. en grundlæggende vekselvirkning. I stedet er det ofte fænomenologiske potentialer, der bruges til formålet, men i artiklen [1] bliver der introduceret en kernemodel, hvor kernepartiklerne er i en superpositionstilstand med mesoner, som behandles eksplicit.

I dette projekt anvendes denne model på den mest simple kerne, deuteronen, i en superpositionstilstand med op til 2 skalar-isoskalar σ -mesoner, hvilket fører til et fire-legemers-problem, der skal løses. Dette gøres numerisk med den korrelerede Gaussiske metode. Beregninger i [1] viser, at der vha. kun en enkelt σ -meson allerede findes en bunden tilstand af deuteronen. Som forventet blev der fundet ud af, at den anden meson giver en ekstra bidrag til bindingsenergien. For rimelige valg af modelparametre, som en mesonmasse på $m_\sigma = 500$ MeV, der giver en bindingsenergi på -2.2 MeV i modellen med en enkelt meson, giver den anden meson et bidrag på -0.173 MeV eller ca. 7% af den samlede bindingsenergi. Der vises, at betydningen af den anden meson stiger, hvis mesonmassen m_σ falder ned til pion massen på $m_\pi = 137$ MeV.

Der vises, at den anden meson giver en lille korrektur på bølgefunktionen, som er opnået ved den korrelerede Gaussiske metode. Metoden giver desuden automatisk en bølgefunktion, der er symmetrisk med henhold til ombytningen af de uskelnelige mesoner.

I forlængelse af projektet kunne effekten af en tredje σ -meson undersøges, hvilket forventes kun at bidrage med 1% eller mindre til deutronens bindingsenergi. Den stigende betydning af den anden meson ved mindre masser peger på, at en kernemodel, der inkluderer mindst to pioner, burde undersøges.

3 Introduction

Models that describe the binding of nuclei using explicit mesons are of particular interest. If they work, the advantage is the naturally implementation of mesonic physics [2] and more generality of the model, since phenomenological potentials are replaced by a few parameters. In [1] a nuclear model is proposed, where the nucleons are emitting and absorbing mesons. The model is applied to the deuteron, since it is the simplest bound nucleon system, consisting of only a proton and a neutron. The investigated exchange meson is *one* σ -meson, also named $f_0(500)$, a light scalar-isoscalar meson candidate [3]. The aim of this thesis is to work out the contribution of a second σ -meson on the deuteron system. The scalar-isoscalar nature makes it easy to work with, since the spin or isospin formalism is not needed in order to describe the interactions in the system.

Whereas the example calculations on the deuteron in [1] work with a two component wave function structure, in this report it is extended to three components; a subsystem with zero, one and two mesons. The same mechanism, W , that couples the first σ -meson to the proton-neutron subsystem, is used to couple the second meson to the three-body subsystem. The corresponding Hamiltonian is then of the form

$$H = \begin{pmatrix} K_N & W_{\sigma_1} & 0 \\ W_{\sigma_1} & K_N + K_{\sigma_1} + m_\sigma & W_{\sigma_2} \\ 0 & W_{\sigma_2} & K_N + K_{\sigma_1} + K_{\sigma_2} + 2m_\sigma \end{pmatrix}, \quad (1)$$

where K_N is the kinetic energy of the nucleon, $K_{\sigma_1}, K_{\sigma_2}$ the kinetic energy of the mesons and m_σ the meson mass. The operators W_{σ_1} and W_{σ_2} generate and annihilate the mesons. The Schrödinger equation to solve is then

$$\begin{pmatrix} K_N & W_{\sigma_1} & 0 \\ W_{\sigma_1} & K_N + K_{\sigma_1} + m_\sigma & W_{\sigma_2} \\ 0 & W_{\sigma_2} & K_N + K_{\sigma_1} + K_{\sigma_2} + 2m_\sigma \end{pmatrix} \begin{pmatrix} \psi_{np} \\ \psi_{\sigma np} \\ \psi_{\sigma\sigma np} \end{pmatrix} = E \begin{pmatrix} \psi_{np} \\ \psi_{\sigma np} \\ \psi_{\sigma\sigma np} \end{pmatrix}, \quad (2)$$

where ψ_{np} , $\psi_{\sigma np}$ and $\psi_{\sigma\sigma np}$ are the wave functions of the three subsystems. [1] already suggests a bound state with just one meson, so this thesis focuses on what contribution to the ground state energy the second meson delivers.

This eigenvalue problem is solved numerically using the correlated Gaussian method, where the wave function is approximated as linear combination of correlated Gaussians. These are created stochastically using pseudo-random van der Corput sampling for the parameters, which ensures fast convergence [4].

In the next sections i) the notation and ii) theory behind the deuteron and the correlated Gaussian method are explained; iii) the numerical results are presented, iv) discussed and v) a conclusion is made.

4 Notation and Coordinates

In this thesis few-body systems are investigated, so properly choice of notation and coordinates should be made. A system consisting of N particles has an associated super vector $\mathbf{r}$, containing the N position vectors,

$$\mathbf{r} = \begin{pmatrix} \vec{r}_1 \\ \vec{r}_2 \\ \vdots \\ \vec{r}_N \end{pmatrix}. \quad (3)$$

Within this notation, vectors $w \in \mathbb{R}^N$ are particular useful, when used to extract position vectors from $\mathbf{r}$. Consider for example w_i with 1 on the i 'th entry and zeros on all other entries. Then,

$$\vec{r}_i = w_i^\top \mathbf{r}. \quad (4)$$

4.1 Coordinate transformation

Suppose a coordinate transformation $\mathbf{r} \rightarrow J\mathbf{r} = \mathbf{x}$, where J is a $N \times N$ matrix. Then $\mathbb{R}^N \ni w \rightarrow U^\top w = v$, where $U = J^{-1}$, since

$$v^\top \mathbf{x} = (U^\top w)^\top J\mathbf{r} = w^\top U J\mathbf{r} = w^\top \mathbf{r} \quad (5)$$

Jacobi coordinates

Particular useful are the Jacobi coordinates, which are the particle's relative positions and the center-of-mass coordinate. They are defined by

$$\vec{x}_{n < N} = \frac{\sum_{i=1}^n m_i \vec{r}_i}{\sum_{i=1}^n m_i} - \vec{r}_{n+1}, \quad \vec{x}_N = \frac{\sum_{i=1}^N m_i \vec{r}_i}{\sum_{i=1}^N m_i}, \quad (6)$$

yielding the transformation matrix

$$J = \begin{pmatrix} 1 & -1 & 0 & 0 & \dots & 0 \\ \frac{m_1}{M_2} & \frac{m_2}{M_2} & -1 & 0 & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ \frac{m_1}{M_{N-1}} & \frac{m_2}{M_{N-1}} & \dots & \dots & \dots & -1 \\ \frac{m_1}{M_N} & \frac{m_2}{M_N} & \dots & \dots & \dots & \frac{m_N}{M_N} \end{pmatrix}, \quad (7)$$

where $M_n = \sum_{i=1}^n m_i$. The last coordinate is the center-of-mass coordinate, which is thrown away, since only intern nuclear interactions are of interest. This is done by cutting off the last row in J and equivalently the last column in J^{-1} . An advantage is that an N -body problem turns into an $N - 1$ -body problem, reducing computational effort.

5 Theory

5.1 Variational Principle

The variational principle is used to find an upper bound for the ground state energy of a quantum mechanical system. The principle states that the expectation value of a system's Hamiltonian $\hat{H}$ for any normalized trial wave function ψ is greater than the ground state energy E_{gs} [5],

$$E_{\text{gs}} \leq \langle \psi | \hat{H} | \psi \rangle. \quad (8)$$

Often, the actual wave function as well the ground state energy are unknown, but the Hamiltonian is easy to find. Then by using different trial wave functions, an educated guess is formed by looking for functions, which minimize the expectation value of the Hamiltonian. In this project the trial wave functions are formed as linear combinations of correlated Gaussians, $|A\rangle$, see Section 5.3. The advantage is that the expectation value in Eq. (8) is found by analytic matrices [6].

5.2 Deuteron as a system of two nucleons and two σ -meson

The spatial wave function of the deuteron in this model is a three components structure,

$$\psi = \begin{pmatrix} \psi_{\text{np}}(\vec{r}_{\text{n}}, \vec{r}_{\text{p}}) \\ \psi_{\sigma\text{np}}(\vec{r}_{\sigma_1}, \vec{r}_{\text{n}}, \vec{r}_{\text{p}}) \\ \psi_{\sigma\sigma\text{np}}(\vec{r}_{\sigma_2}, \vec{r}_{\sigma_1}, \vec{r}_{\text{n}}, \vec{r}_{\text{p}}) \end{pmatrix}. \quad (9)$$

ψ_{np} contains only the proton and the neutron. $\psi_{\sigma\text{np}}$ is the wave function of the three-body subsystem, where the nucleus is dressed with one σ -meson and $\psi_{\sigma\sigma\text{np}}$ the corresponding subsystem with two mesons. The associated Hamiltonian is a matrix,

$$\hat{H} = \begin{pmatrix} \hat{K}_{\text{n}} + \hat{K}_{\text{p}} & \hat{W}_{\sigma_1} & 0 \\ \hat{W}_{\sigma_1} & \hat{K}_{\text{n}} + \hat{K}_{\text{p}} + \hat{K}_{\sigma_1} + m_{\sigma} & \hat{W}_{\sigma_2} \\ 0 & \hat{W}_{\sigma_2} & \hat{K}_{\text{n}} + \hat{K}_{\text{p}} + \hat{K}_{\sigma_1} + \hat{K}_{\sigma_2} + 2m_{\sigma} \end{pmatrix}, \quad (10)$$

where $\hat{K}_{\text{n}}$, $\hat{K}_{\text{p}}$, $\hat{K}_{\sigma_1}$ and $\hat{K}_{\sigma_2}$ are the kinetic energy operators for the neutron, proton and the mesons and m_{σ} is the σ -meson's mass. $\hat{W}_{\sigma_1}$ and $\hat{W}_{\sigma_2}$ are the meson-generating/annihilation operators, which are defined later.

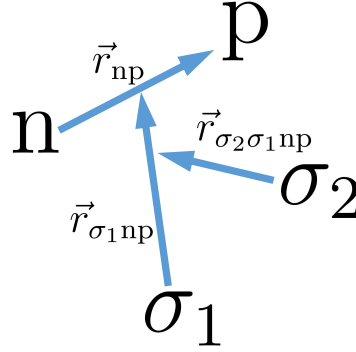


Figure 1 A set of Jacobi coordinates for the deuteron- σ - σ subsystem.

Kinetic energy operator

The problem is easier to handle by using Jacobi coordinates, defined in Section 4.1. Since the last coordinate is the center-of-mass coordinate, it can be omitted, as only the internuclear dynamics are explored. The Jacobi coordinates get,

$$\vec{r}_{np} = \vec{r}_p - \vec{r}_n, \quad \vec{r}_{\sigma_1 np} = \frac{m_n \vec{r}_n + m_p \vec{r}_p}{m_n + m_p} - \vec{r}_{\sigma_1}, \quad \vec{r}_{\sigma_2 \sigma_1 np} = \frac{m_n \vec{r}_n + m_p \vec{r}_p + m_\sigma \vec{r}_{\sigma_1}}{m_n + m_p + m_\sigma} - \vec{r}_{\sigma_2}, \quad (11)$$

see Fig. 1 for illustration.

With the reduced masses,

$$\mu_{np} = \frac{m_n m_p}{m_n + m_p}, \quad \mu_{\sigma np} = \frac{m_\sigma (m_n + m_p)}{m_n + m_p + m_\sigma}, \quad \mu_{\sigma \sigma np} = \frac{m_\sigma (m_n + m_p + m_\sigma)}{m_n + m_p + 2m_\sigma}, \quad (12)$$

and the corresponding relative kinetic energy operators¹,

$$\hat{K}_{np} = -\frac{\hbar^2}{2\mu_{np}} \frac{\partial^2}{\partial \vec{r}_{np}^2}, \quad \hat{K}_{\sigma_1 np} = -\frac{\hbar^2}{2\mu_{\sigma np}} \frac{\partial^2}{\partial \vec{r}_{\sigma_1 np}^2}, \quad \hat{K}_{\sigma_2 \sigma_1 np} = -\frac{\hbar^2}{2\mu_{\sigma \sigma np}} \frac{\partial^2}{\partial \vec{r}_{\sigma_2 \sigma_1 np}^2}, \quad (13)$$

the Hamiltonian in the center-of-mass systems becomes,

$$\hat{H} = \begin{pmatrix} \hat{K}_{np} & \hat{W}_{\sigma_1} & 0 \\ \hat{W}_{\sigma_1} & \hat{K}_{np} + \hat{K}_{\sigma_1 np} + m_\sigma & \hat{W}_{\sigma_2} \\ 0 & \hat{W}_{\sigma_2} & \hat{K}_{np} + \hat{K}_{\sigma_1 np} + \hat{K}_{\sigma_2 \sigma_1 np} + 2m_\sigma \end{pmatrix}. \quad (14)$$

The Schrödinger equation then reads,

¹Non-relativistic kinetic energy operators. Due to the mass terms, m_σ , in the Hamiltonian, the σ -mesons are under barrier and are not expected to move with relativistic speed.

$$\hat{H} \begin{pmatrix} \psi_{np}(\vec{r}_{np}) \\ \psi_{\sigma np}(\vec{r}_{np}, \vec{r}_{\sigma 1 np}) \\ \psi_{\sigma \sigma np}(\vec{r}_{np}, \vec{r}_{\sigma 1 np}, \vec{r}_{\sigma 2 \sigma 1 np}) \end{pmatrix} = E \begin{pmatrix} \psi_{np}(\vec{r}_{np}) \\ \psi_{\sigma np}(\vec{r}_{np}, \vec{r}_{\sigma 1 np}) \\ \psi_{\sigma \sigma np}(\vec{r}_{np}, \vec{r}_{\sigma 1 np}, \vec{r}_{\sigma 2 \sigma 1 np}) \end{pmatrix}. \quad (15)$$

The coupling operator

The coupling operator for the first meson is introduced on integral form,

$$\langle \psi_{np} | \hat{W}_{\sigma 1} | \psi_{\sigma 1 np} \rangle = \int d^3 \vec{r}_{np} d^3 \vec{r}_{\sigma 1 np} \psi_{np}^*(\vec{r}_{np}) \hat{W}_{\sigma 1}(\vec{r}_{np}, \vec{r}_{\sigma 1 np}) \psi_{\sigma 1 np}(\vec{r}_{np}, \vec{r}_{\sigma 1 np}), \quad (16)$$

where $\hat{W}_{\sigma 1}(\vec{r}_{np}, \vec{r}_{\sigma 1 np})$ has to be short range, i.e. has to vanish when either proton, neutron or σ -meson are at larger distances [1]. Here, this is achieved by a Gaussian,

$$\hat{W}_{\sigma 1}(\vec{r}_{np}, \vec{r}_{\sigma 1 np}) = S_{\sigma} \exp \left(-\frac{\vec{r}_{np}^2 + \vec{r}_{\sigma 1 np}^2}{b_{\sigma}^2} \right), \quad (17)$$

where the range b_{σ} and strength S_{σ} are parameters of the model.

This is just the operator that couples the subsystem with no mesons to the subsystem with one meson. The operator $\hat{W}_{\sigma 2}$ that generates/annihilates a second meson must be on the same form, with $\vec{r}_{\sigma 1 np} \rightarrow \vec{r}_{\sigma 2 np}$,

$$\hat{W}_{\sigma 2}(\vec{r}_{np}, \vec{r}_{\sigma 2 np}) = S_{\sigma} \exp \left(-\frac{\vec{r}_{np}^2 + \vec{r}_{\sigma 2 np}^2}{b_{\sigma}^2} \right), \quad (18)$$

where²,

$$\vec{r}_{\sigma 2 np} = \frac{m_n \vec{r}_n + m_p \vec{r}_p}{m_n + m_p} - \vec{r}_{\sigma 2} = \vec{r}_{\sigma 1 np} + \vec{r}_{\sigma 1} - \vec{r}_{\sigma 2}. \quad (19)$$

5.3 Correlated Gaussian Method

In this method, correlated Gaussians, form a (non-orthonormal³) basis for the spatial wave functions,

$$\psi = \sum_{i=1} c_i \langle \mathbf{r} | A_i \rangle. \quad (20)$$

²This belongs to a second set of Jacobi coordinates, where the order between the first and second mesons is exchanged. Since the mesons are of equal mass, the transformation from the n,p, σ_1 , σ_2 -coordinates is given by the matrix J' , which is equivalent with exchanging the last two columns in the transformation matrix J , belonging to the set of coordinates (11).

³ $\langle A' | A \rangle \neq 0$, for $|A\rangle$ and $|A'\rangle$ correlated Gaussians. See Eqs. (31)-(33)

A correlated Gaussian is defined as,

$$\langle \mathbf{r} | A \rangle = \exp \left(- \sum_{i,j=1}^N (A)_{ij} \vec{r}_i \cdot \vec{r}_j \right) \equiv e^{-\mathbf{r}^\top A \mathbf{r}}, \quad (21)$$

where the N -size column vector $\mathbf{r} = (\vec{r}_1, \vec{r}_2, \dots, \vec{r}_N)^\top$ is the collection of the system's coordinates, see **Section 4**, and A is a $N \times N$ -matrix of parameters.

The linear combination can be inserted into the Schrödinger equation,

$$\hat{H} \sum_{i=1} c_i |A_i\rangle = E \sum_{i=1} c_i |A_i\rangle, \quad (22)$$

and multiplied by $\langle A_j |$ from the left, to yield,

$$\sum_{i=1} c_i \langle A_j | \hat{H} | A_i \rangle = E \sum_{i=1} c_i \langle A_j | A_i \rangle. \quad (23)$$

This can be rewritten as a generalized eigenvalue problem,

$$\mathcal{H} \mathbf{c} = E \mathcal{N} \mathbf{c}, \quad (24)$$

where $\mathbf{c}$ is a column vector consisting of the linear c_i -coefficients, $\mathcal{H}_{ij} = \langle A_j | \hat{H} | A_i \rangle$ and $\mathcal{N}_{ij} = \langle A_j | A_i \rangle$. These matrix elements are derived in the next sections.

The deuteron system

The subsystems of the deuteron system with up to two mesons are described by their Jacobi coordinates, defined in Eq. (11). By omitting the center-of-mass coordinate, a one-component structure for the two-body np-subsystem,

$$\mathbf{r}^{(d)} = (\vec{r}_{\text{np}}), \quad (25)$$

is obtained, so the parameter matrices $\{A_i^{(d)}\}$ are one-time-one matrices – the matrix formalism is kept for consistency. The mesons subsystems have the coordinates-vectors,

$$\mathbf{r}^{(\sigma)} = \begin{pmatrix} \vec{r}_{\text{np}} \\ \vec{r}_{\sigma_1 \text{np}} \end{pmatrix}, \quad \mathbf{r}^{(2\sigma)} = \begin{pmatrix} \vec{r}_{\text{np}} \\ \vec{r}_{\sigma_1 \text{np}} \\ \vec{r}_{\sigma_2 \sigma_1 \text{np}} \end{pmatrix}, \quad (26)$$

yielding 2×2 parameter matrices $\{A_j^{(\sigma)}\}$ for the σ np-subsystem and 3×3 parameter matrices $\{A_k^{(2\sigma)}\}$ for the 2 mesons subsystem. This leads to the wave function representations,

$$\begin{aligned}
\psi_{\text{np}}(\mathbf{r}^{(\text{d})}) &= \sum_{i=1}^{n^{(\text{d})}} c_i^{(\text{d})} e^{-\mathbf{r}^{(\text{d})\text{T}} A_i^{(\text{d})} \mathbf{r}^{(\text{d})}} \equiv \sum_{i=1}^{n^{(\text{d})}} c_i^{(\text{d})} \langle \mathbf{r}^{(\text{d})} | A_i^{(\text{d})} \rangle, \\
\psi_{\sigma\text{np}}(\mathbf{r}^{(\sigma)}) &= \sum_{j=1}^{n^{(\sigma)}} c_j^{(\sigma)} e^{-\mathbf{r}^{(\sigma)\text{T}} A_j^{(\sigma)} \mathbf{r}^{(\sigma)}} \equiv \sum_{j=1}^{n^{(\sigma)}} c_j^{(\sigma)} \langle \mathbf{r}^{(\sigma)} | A_j^{(\sigma)} \rangle, \\
\psi_{\sigma\sigma\text{np}}(\mathbf{r}^{(2\sigma)}) &= \sum_{k=1}^{n^{(2\sigma)}} c_k^{(2\sigma)} e^{-\mathbf{r}^{(2\sigma)\text{T}} A_k^{(2\sigma)} \mathbf{r}^{(2\sigma)}} \equiv \sum_{k=1}^{n^{(2\sigma)}} c_k^{(2\sigma)} \langle \mathbf{r}^{(2\sigma)} | A_k^{(2\sigma)} \rangle,
\end{aligned} \tag{27}$$

where $n^{(\text{d})}$, $n^{(\sigma)}$ and $n^{(2\sigma)}$ are the number of Gaussians in the three subsystems.⁴ The non-linear parameters $\{A_i^{(\text{d})}, A_j^{(\sigma)}, A_k^{(2\sigma)}\}$ are chosen stochastically, see Section 5.4, while the linear parameters $\{c_i^{(\text{d})}, c_j^{(\sigma)}, c_k^{(2\sigma)}\}$ are found by solving the generalized eigenvalue problem, Eq. (24).

The Hamiltonian matrix $\mathcal{H}$ is of the following block matrix structure,

$$\begin{pmatrix}
\langle A_i^{(\text{d})} | \hat{K}_{\text{np}} | A_{i'}^{(\text{d})} \rangle & \langle A_i^{(\text{d})} | \hat{W}_{\sigma_1} | A_{j'}^{(\sigma)} \rangle & 0 \\
\langle A_j^{(\sigma)} | \hat{W}_{\sigma_1} | A_i^{(\text{d})} \rangle & \langle A_j^{(\sigma)} | \hat{K}^{(\sigma)} + m_\sigma | A_{j'}^{(\sigma)} \rangle & \langle A_j^{(\sigma)} | \hat{W}_{\sigma_2} | A_k^{(2\sigma)} \rangle \\
0 & \langle A_k^{(2\sigma)} | \hat{W}_{\sigma_2} | A_j^{(\sigma)} \rangle & \langle A_k^{(2\sigma)} | \hat{K}^{(2\sigma)} + 2m_\sigma | A_{k'}^{(2\sigma)} \rangle
\end{pmatrix}, \tag{28}$$

where $\hat{K}^{(\sigma)} = \hat{K}_{\text{np}} + \hat{K}_{\sigma_1\text{np}}$ and $\hat{K}^{(2\sigma)} = \hat{K}_{\text{np}} + \hat{K}_{\sigma_1\text{np}} + \hat{K}_{\sigma_2\sigma_1\text{np}}$. Equivalently the overlap matrix $\mathcal{N}$ becomes a 3×3 block matrix too,

$$\mathcal{N} = \begin{pmatrix}
\langle A_i^{(\text{d})} | A_{i'}^{(\text{d})} \rangle & 0 & 0 \\
0 & \langle A_j^{(\sigma)} | A_{j'}^{(\sigma)} \rangle & 0 \\
0 & 0 & \langle A_k^{(2\sigma)} | A_{k'}^{(2\sigma)} \rangle
\end{pmatrix}. \tag{29}$$

The indices in both matrices are $i, i' = 1, 2, \dots, n^{(\text{d})}$, $j, j' = 1, 2, \dots, n^{(\sigma)}$ and $k, k' = 1, 2, \dots, n^{(2\sigma)}$.

The correlated Gaussian method has the advantage that the matrix elements in Eqs. (28) and (29) are analytic. They are derived in the next sections. The cross terms are zero,

$$\langle A^{(a)} | \hat{X} | A^{(b)} \rangle = 0, \quad a, b = \text{d}, \sigma, 2\sigma, \quad a \neq b, \tag{30}$$

for all operators $\hat{X}$, except of the coupling operators $\hat{W}_{\sigma_1}$ and $\hat{W}_{\sigma_2}$.

⁴These numbers are chosen conveniently; increasing the number of Gaussians results in more accurate approximations to the wave function, but is of course accompanied with more computational load. See the convergence plots, Fig. 6.

Overlap

The overlap between two correlated Gaussians, $|A\rangle$ and $|A'\rangle$, is the integral

$$\langle A'|A\rangle = \int d^3\vec{r}_1 d^3\vec{r}_2 \dots d^3\vec{r}_N e^{-\mathbf{r}^\top A' \mathbf{r}} e^{-\mathbf{r}^\top A \mathbf{r}} = \int d^3\vec{r}_1 d^3\vec{r}_2 \dots d^3\vec{r}_N e^{-\mathbf{r}^\top (A'+A) \mathbf{r}}. \quad (31)$$

Denote $B = A' + A$ and consider an orthogonal coordinate transformation $\mathbf{r} = Q\mathbf{x}$, to the basis, where matrix B is diagonal. Then $B = QDQ^\top$, where D is a diagonal matrix and $Q^\top Q = QQ^\top = \mathbf{1}$.⁵ Due to the fact that D is diagonal, it follows,

$$e^{-(Q\mathbf{x})^\top B Q\mathbf{x}} = e^{-\mathbf{x}^\top D \mathbf{x}} = e^{-\sum_{i=1}^N \bar{x}_i D_{ii} \bar{x}_i} = \prod_{i=1}^N e^{-D_{ii} \bar{x}_i^2}, \quad (32)$$

where $x_i = \|\vec{x}_i\|$. This gives Gaussian integrals, which result in⁶

$$\prod_{i=1}^N \int d^3\vec{x}_i e^{-D_{ii} x_i^2} = \prod_{i=1}^N \int d\Omega_i \int_0^\infty dx_i x_i^2 e^{-D_{ii} x_i^2} = \prod_{i=1}^N \left(\frac{\pi}{D_{ii}} \right)^{3/2} = \left(\frac{\pi^N}{\det B} \right)^{3/2}. \quad (33)$$

Kinetic Energy

The non-relativistic kinetic energy operator of an N -body system is equal to

$$\hat{K} = - \sum_{i=1}^N \frac{\hbar^2}{2m_i} \frac{\partial^2}{\partial \vec{r}_i^2}. \quad (34)$$

The matrix K can be introduced, to redefine Eq. (34),

$$\hat{K} = - \sum_{i,j=1}^N \frac{\partial}{\partial \vec{r}_i} K_{ij} \frac{\partial}{\partial \vec{r}_j} \equiv - \frac{\partial}{\partial \mathbf{r}} K \frac{\partial}{\partial \mathbf{r}^\top}. \quad (35)$$

The matrix element becomes [6],

$$\langle A' | \hat{K} | A \rangle = \langle A' | - \frac{\partial}{\partial \mathbf{r}} K \frac{\partial}{\partial \mathbf{r}^\top} | A \rangle = 6 \langle A' | A \rangle \text{trace} \left(A' K A (A + A')^{-1} \right), \quad (36)$$

⁵Since matrices A and A' are symmetric, B is symmetric too. From the Spectral Theorem it follows that the diagonal matrix D and the orthogonal matrix Q exist. See for example Corollary 14.14 in [7].

⁶Last equality comes from

$$\begin{aligned} \det(D) &= \det(Q^\top B Q) = \det(Q^\top) \det(B) \det(Q) = \det(Q^{-1}) \det(B) \det(Q) \\ &= \frac{1}{\det(Q)} \det(B) \det(Q) = \det(B) \end{aligned}$$

where the K -matrices for the three subsystems are given as,

$$K^{(d)} = \begin{pmatrix} \frac{\hbar^2}{2\mu_{np}} \end{pmatrix}, \quad K^{(\sigma)} = \begin{pmatrix} \frac{\hbar^2}{2\mu_{np}} & 0 \\ 0 & \frac{\hbar^2}{2\mu_{\sigma np}} \end{pmatrix}, \quad K^{(2\sigma)} = \begin{pmatrix} \frac{\hbar^2}{2\mu_{np}} & 0 & 0 \\ 0 & \frac{\hbar^2}{2\mu_{\sigma np}} & 0 \\ 0 & 0 & \frac{\hbar^2}{2\mu_{\sigma\sigma np}} \end{pmatrix}. \quad (37)$$

Coupling operators

The coupling operator $\hat{W}_{\sigma_1}$ that annihilates/generates the first meson, can be represented in the correlated Gaussian basis and the matrix element becomes,

$$\langle A^{(d)} | \hat{W}_{\sigma_1} | A^{(\sigma)} \rangle = \langle A^{(d)} | S_\sigma \exp \left(-\frac{\vec{r}_{np}^2 + \vec{r}_{\sigma_1 np}^2}{b_\sigma^2} \right) | A^{(\sigma)} \rangle = S_\sigma \langle \tilde{A}^{(d)} | A^{(\sigma)} \rangle, \quad (38)$$

with,

$$\tilde{A}^{(d)} = \begin{pmatrix} A^{(d)} & 0 \\ 0 & 0 \end{pmatrix} + \frac{w_1 w_1^T + w_2 w_2^T}{b_\sigma^2} = \begin{pmatrix} A^{(d)} + \frac{1}{b_\sigma^2} & 0 \\ 0 & \frac{1}{b_\sigma^2} \end{pmatrix}, \quad (39)$$

where $w_i \in \mathbb{R}^2$ is a column vector with 1 on the i 'th entry and 0 on the other(s), see Eq. (4).

For the second operator, $\hat{W}_{\sigma_2}$, the same technique is used, but due to the $\vec{r}_{\sigma_2 np}$ -coordinate in Eq. (18), a coordinate transformation, T , is needed⁷,

$$\mathbf{r}'^{(2\sigma)} = \begin{pmatrix} \vec{r}_{np} \\ \vec{r}_{\sigma_2 np} \\ \vec{r}_{\sigma_1 \sigma_2 np} \end{pmatrix} \rightarrow \mathbf{r}^{(2\sigma)} = \begin{pmatrix} \vec{r}_{np} \\ \vec{r}_{\sigma_1 np} \\ \vec{r}_{\sigma_2 \sigma_1 np} \end{pmatrix} = T \mathbf{r}'^{(2\sigma)}, \quad (40)$$

where $\vec{r}_{\sigma_2 np}$ is defined in Eq. (19) and

$$\vec{r}_{\sigma_1 \sigma_2 np} = \frac{m_n \vec{r}_n + m_p \vec{r}_p + m_\sigma \vec{r}_{\sigma_2}}{m_n + m_p + m_\sigma} - \vec{r}_{\sigma_1}, \quad (41)$$

see Fig. 2, for illustration of the new Jacobi coordinates, compared to Fig. 1.

The matrix element becomes,

$$\langle A^{(\sigma)} | \hat{W}_{\sigma_2} | A^{(2\sigma)} \rangle = \langle A^{(\sigma)} | S_\sigma \exp \left(-\frac{\vec{r}_{np}^2 + \vec{r}_{\sigma_2 np}^2}{b_\sigma^2} \right) | A^{(2\sigma)} \rangle = S_\sigma \langle \tilde{A}^{(\sigma)} | A^{(2\sigma)} \rangle, \quad (42)$$

⁷From the discussion in footnote 2, p. 8, it follows with the footnote's definitions that $T = J J'^{-1}$.

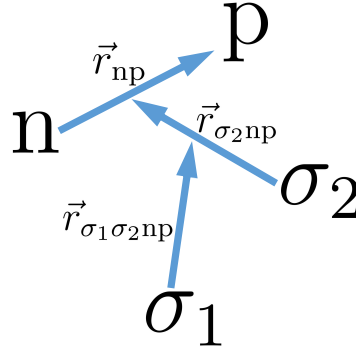


Figure 2 A second set of Jacobi coordinates for the deuteron- σ - σ subsystem, which is different from Fig. 1.

where,

$$\tilde{A}^{(\sigma)} = \begin{pmatrix} A^{(\sigma)} & 0 \\ 0 & 0 & 0 \end{pmatrix} + \frac{(T^{-1})^T (w_1 w_1^T + w_2 w_2^T) T^{-1}}{b_\sigma^2}, \quad (43)$$

with the familiar column vectors $w_i \in \mathbb{R}^3$, which transform $w_i \rightarrow (T^{-1})^T w_i$ under (40), see Eq. (5) and Section 4.

Symmetry of the wave function

Since the two σ -mesons are identical bosons, the wave function must be symmetric under the exchange of meson positions, $\vec{r}_{\sigma_1} \leftrightarrow \vec{r}_{\sigma_2}$, carried out by the exchange operator, $\hat{S}$. If the wave function that solves the Schrödinger equation (2), is symmetric, then the Schrödinger equation can be written as,

$$\hat{S} \hat{H} \hat{S}^{-1} \hat{S} \psi = \hat{S} E \psi. \quad (44)$$

It can be concluded that $\hat{S} \hat{H} \hat{S}^{-1} = \hat{H}$, since the way the Hamiltonian (1) is formed, it will under the operation $\hat{S}$ be equal to the same Hamiltonian operating on $\hat{S} \psi$.⁸

In $\mathbf{r} = (\vec{r}_n, \vec{r}_p, \vec{r}_{\sigma_1}, \vec{r}_{\sigma_2})^T$ -coordinates, the exchange operator is given by the matrix,

$$S = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}. \quad (45)$$

⁸This becomes more clear in the Matrix representation of the Hamiltonian in the basis of correlated Gaussians, Eq. (28), where the operation $\hat{S}$ on $\mathcal{H}$ is equivalent with the operation $\hat{S}$ on the correlated Gaussians, $|A\rangle$.

In the same coordinates, the wave function in the basis of correlated Gaussians (21), transform like,

$$\hat{S}\psi = \hat{S} \sum_i c_i \langle \mathbf{r} | A_i \rangle = \sum_i c_i e^{\mathbf{r}^\top S^\top A_i S \mathbf{r}} = \sum_i c_i \langle \mathbf{r} | S^\top A_i S \rangle, \quad (46)$$

so the c_i -coefficients should stay the same, if the wave function is symmetric.

5.4 Stochastic Parameter Sampling

The correlated Gaussians for an N -body system are parametrized in the form [1],

$$\langle \mathbf{r} | A \rangle = \exp \left(- \sum_{i < j=1}^N \left(\frac{\vec{r}_i - \vec{r}_j}{b_{ij}} \right)^2 \right) \equiv \exp(\mathbf{r}^\top A \mathbf{r}). \quad (47)$$

In practice the matrix A is build,

$$A = \sum_{i < j=1}^N \frac{w_{ij} w_{ij}^\top}{b^2}, \quad (48)$$

using $w_{ij} \in \mathbb{R}^N$, where the i 'th and j 'th entries are 1 and -1 , and the rest are zero. Under a coordinate transformation, i.e. to Jacobi coordinates, the vectors transform according to Eq. (5).

The parameter b_{ij} is chosen stochastically, with an exponential distribution,

$$b_{ij} = -\ln(u) b_\sigma, \quad (49)$$

where b_σ is the model's range parameter and $u \in]0, 1[$ is taken from a Van der Corput sequence, defined below. For each w_{ij} -vector a separate Van der Corput sequence with different bases in the prime numbers are used. Using quasi-random sampling for u via Van der Corput sequences, instead of a pseudo-random number generator, leads to faster convergence⁹ of the eigenenergies to the ground state energy. The low discrepancy of the quasi-random parameters leads to a thoroughly sampled parameter space, whereas pseudo-random sequences tend to produce relatively large unsampled sub-areas [4].

A Van der Corput sequence is build as follows [4]; a natural number, b , is chosen as base. Then the sequence of natural numbers in base- b representation is,

$$n = \sum_{k=0}^{L-1} d_k(n) b^k, \quad (50)$$

⁹With the same number of pseudo-random and quasi-random sampled Gaussians, the latter will yield a lower eigenenergy, see Fig. 3 in [4].

where $d_k(n) < b$ are the digits and L the “length”, i.e. how many digits are needed to represent the number n . The n 'th number q_n in the Van der Corput sequence is,

$$q_n = \sum_{k=0}^{L-1} d_k(n) b^{-k-1}. \quad (51)$$

6 Results

In the following section the results are presented, where the generalized eigenvalue problem, Eq. (24) is solved in `python` using `scipy.linalg.eigh()`, which is based on LAPACK routines [8].

6.1 Deuteron with one σ -meson

First of all the results in [1] could be reproduced, with the same choice of parameters, where $S_\sigma = 20.35$ MeV and $b_\sigma = 3$ fm. The eigenenergy converges to the experimental binding energy of -2.2 MeV [9] within 10 Gaussians for the np-subsystem and 30 Gaussians for the σ np-subsystem, see Fig. 3.

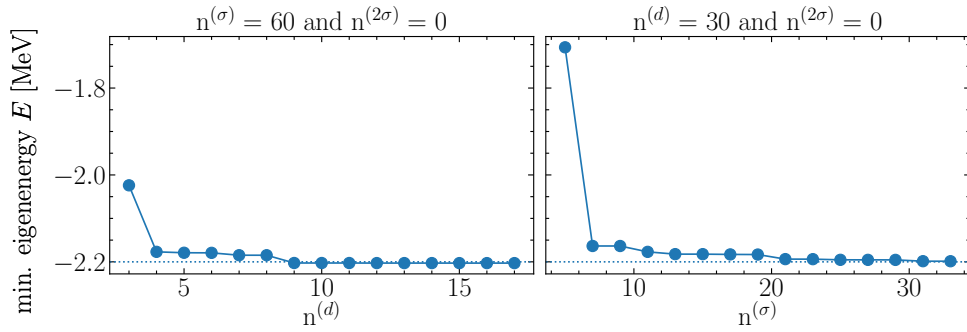


Figure 3 Left: Convergence of the deuteron's ground state energy using $n^{(d)}$ Gaussians for the deuteron subsystem and 60 Gaussians for the 1-meson subsystem. Right: Convergence of the deuteron's ground state energy with 30 Gaussians in the deuteron subsystem and $n^{(\sigma)}$ Gaussians in the deuteron-1-meson subsystem.

Additionally other parameter combinations were investigated. Ground state energy computations, with varying the parameters b_σ , m_σ and S_σ result in Fig. 4. Despite that decreasing the meson mass from 500 MeV to 50 MeV result in lower parameters b_σ and S_σ to yield eigenenergy -2.2 MeV, the opposite is the case below $m_\sigma = 50$ MeV. For $m_\sigma \rightarrow 0$ the parameters b_σ, S_σ tend to grow. To rule out numerical issues, convergence is investigated for the extreme case, see Fig. 5. The model still gives convergence of the energy, with the chosen $n^{(d)} = 30$ and $n^{(\sigma)} = 60$.

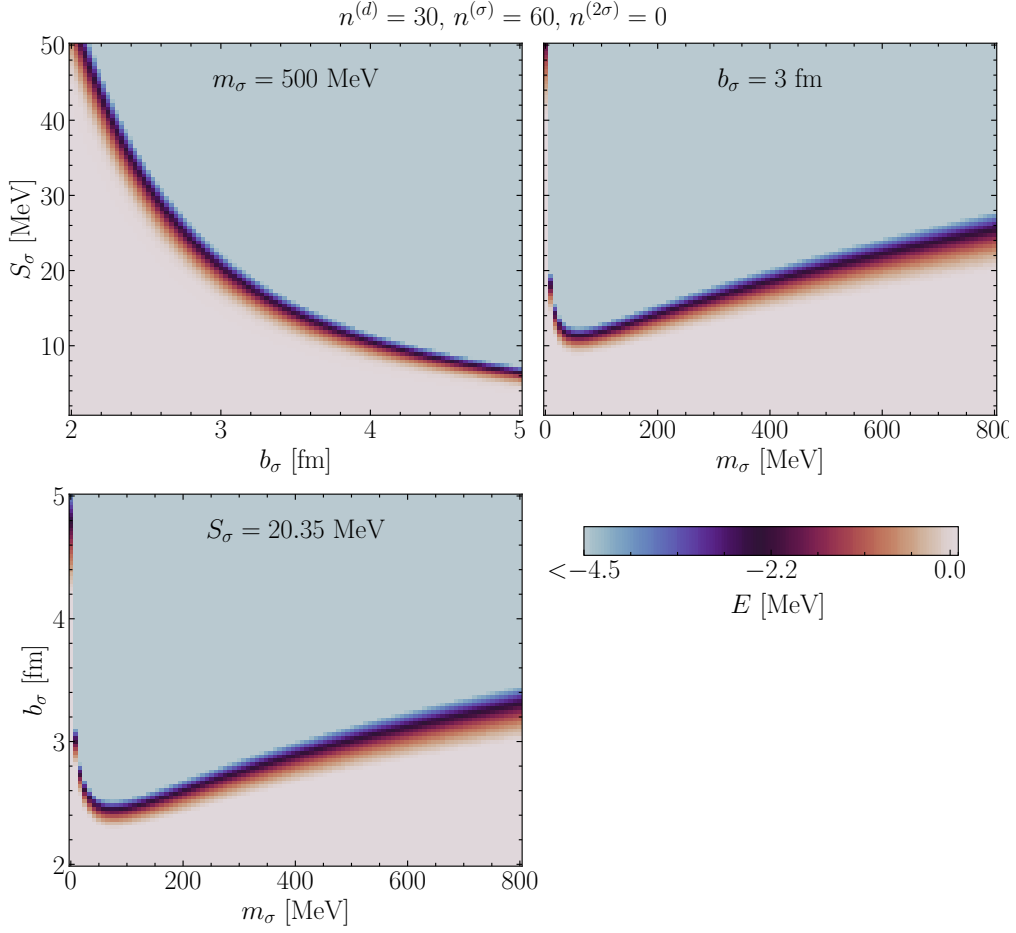


Figure 4 Eigenenergy calculations with one meson for different combinations of the model parameters b_σ , m_σ and S_σ . 30 Gaussians were used for the np-subsystem and 60 Gaussians in the σ np-subsystem.

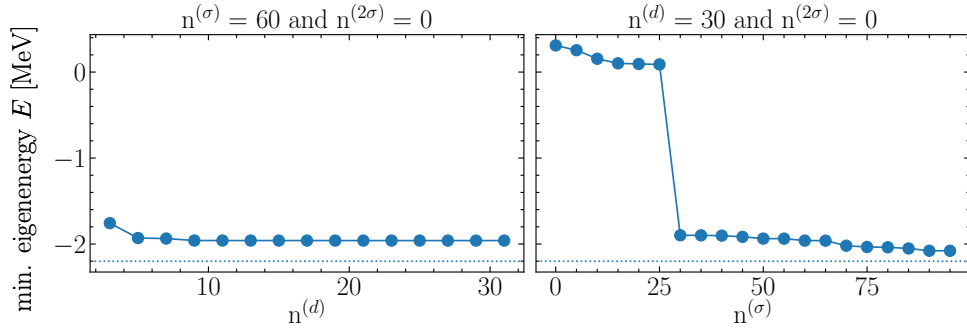


Figure 5 Computations with model parameters $m_\sigma = 1$ MeV and $S_\sigma = 50$ MeV. Left: Convergence of the deuteron's ground state energy using $n^{(d)}$ Gaussians for the deuteron subsystem and 60 Gaussians for the 1-meson subsystem. Right: Convergence of the deuteron's ground state energy with 30 Gaussians in the deuteron subsystem and $n^{(\sigma)}$ Gaussians in the deuteron-1-meson subsystem.

6.2 Deuteron with two σ -mesons

With the same choice of parameters as in [1], see Section 6.1, convergence with increasing number of Gaussians for the deuteron-2-meson subsystem was investigated, see Fig. 6. This resulted in an energy correction of about -0.173 MeV or 7% of the total binding energy. Remarkable is that about 5 Gaussians are enough to let the eigenenergy converge.

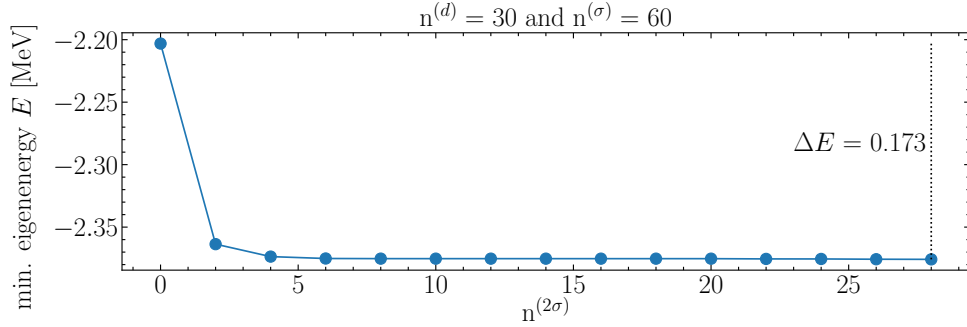


Figure 6 Convergence of the deuteron's ground state energy using $n^{(2\sigma)}$ Gaussians for the 2- σ -np subsystem, 30 Gaussian for the np-subsystem 60 Gaussians for the 1-meson-np subsystem.

The model parameters were varied too, leading to Fig. 7, where the difference in energy compared to the one-meson model is shown. The dashed lines are parameter combinations fitted to $E = -2.200(1)$ MeV in the two-meson model. This shows that the second meson's contribution differs for different parameter combinations. For example the contribution tends to decrease for high and very low masses.

Interesting is the second meson's contribution plotted for the case of varying meson mass, m_σ , and fitted strength parameter, S_σ , on Fig. 8. The energy difference due to the second meson increases until a mass of about 80 MeV and then decreases again, but does not reach zero.

The linear coefficients, $\mathbf{c}$, obtained by solving Eq. (24) are used to construct the wave function for the deuteron, via the linear combination Eq. (20). Only the ψ_{np} -part of the overall wave function is used, since the proton-neutron distance is of interest with highly will be represented by this part [1]. The radial wave function resulting from the parameters $b = 3$ fm and $S_\sigma = 20.35$ MeV is shown on Fig. 9 for both 1 and 2 mesons with mass of 500 MeV. The second meson gives a small correction, which subtracts a small part from the tail and adds a fraction up in the short range, of the normalized radial wave function, $r\psi_{np}^{\vec{r}}$.

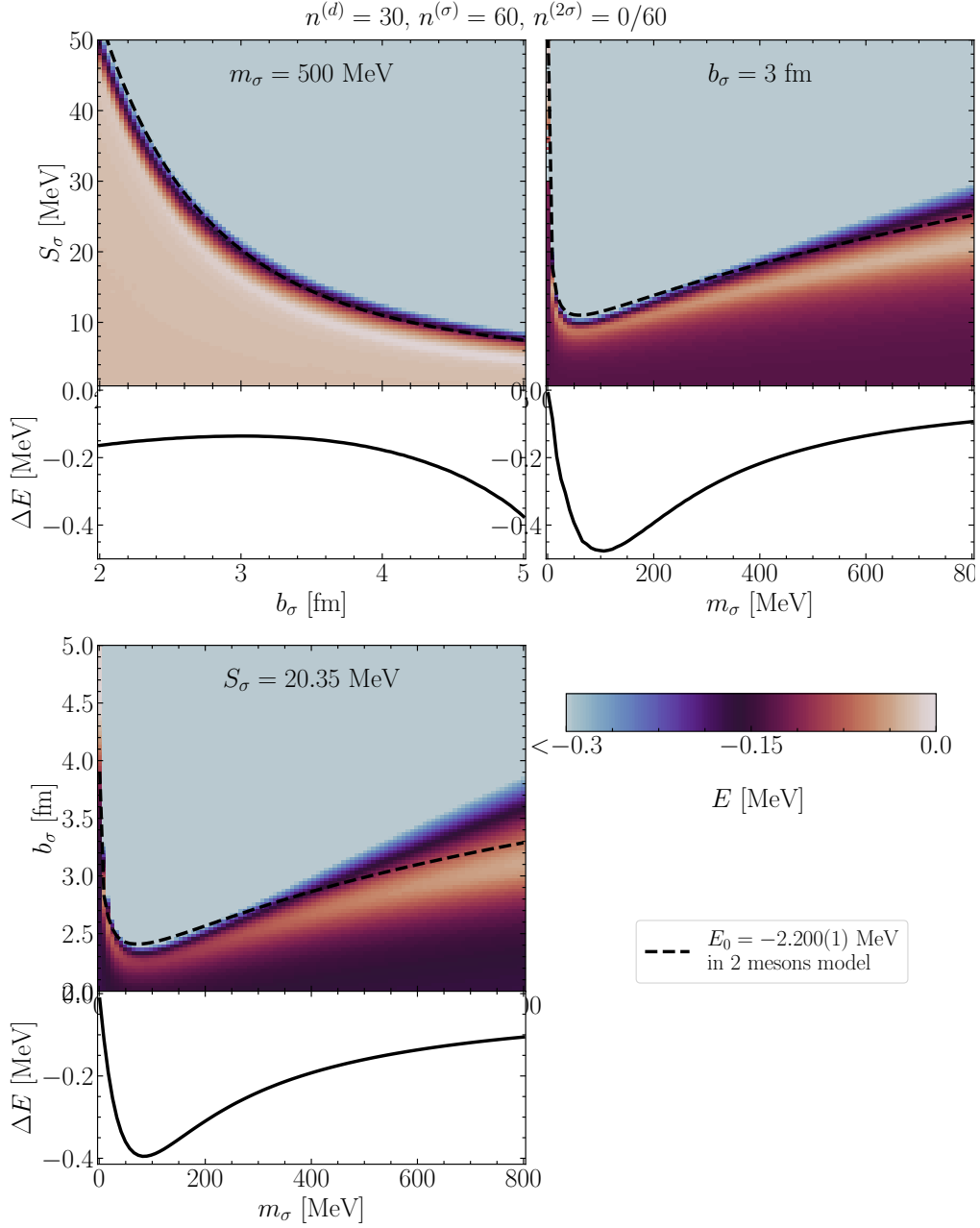


Figure 7 On the contour plots: Difference in ground state energy between computations with one and two mesons, for different combinations of the model parameters b_σ , m_σ and S_σ . 30 Gaussians were used for the np-subsystem and 60 Gaussians in both the σ np-subsystem and $\sigma\sigma$ np-subsystem. Below the contour plots: The difference in energy, due to the second meson with the fitted parameters that yield a groundstate energy of $-2.200(1)$ MeV.

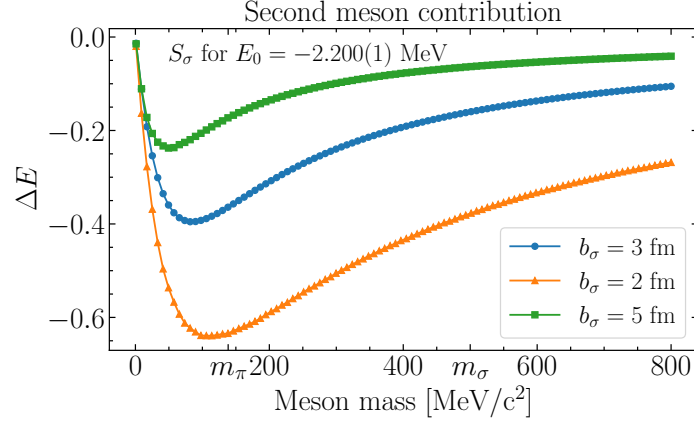


Figure 8 Contribution of second meson to the binding energy in the deuteron for meson masses $m_\sigma \in [1 - 800 \text{ MeV}]$, $b_\sigma = 2, 3, 5 \text{ fm}$ and S_σ was fitted to a ground state energy of $-2.200(1) \text{ MeV}$ with two meson: $n(d) = 30$, $n(\sigma) = 60$, $n(2\sigma)$.

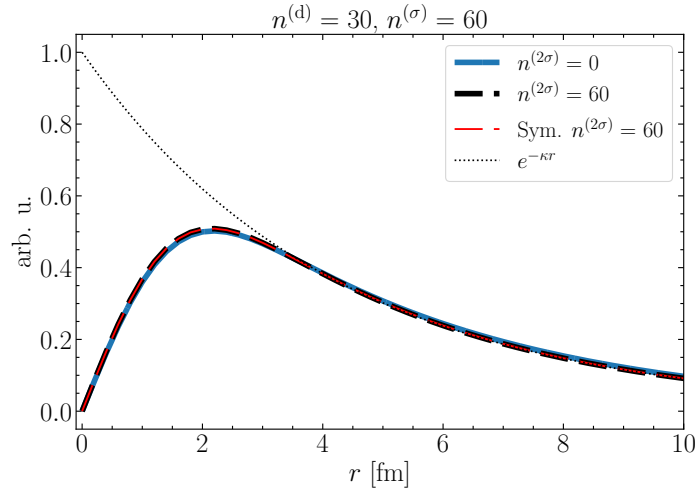


Figure 9 Radial wave function $u(r) = r\psi_{np}(r)$, $r = |\vec{r}_{np}|$, of the proton-neutron subsystem, computed using both 1 (blue) and 2 mesons (black) and for the case where the two meson's positions were switched (red). All wave functions are normalized, so $\int_0^\infty |u(r)|^2 dr = 1$.

7 Discussion

The starting point of in the study of the σ -meson-deuteron model was the three body system consisting the deuteron and one σ -meson. The 1 meson computations with varying range b_σ , strength S_σ and meson mass m_σ , that lead to Fig. 4, show that on a large scale of parameter variation it is possible to yield a ground state energy of -2.2 MeV. However the range of parameters to produce the right energy is small – either the deuteron is not bound at all or the binding energy largely exceeds -2.2 MeV for small deviations from the optimal parameter constellations.

The non-monotonic dependence in the meson mass is of particular interest. There is a need for increasing range b_σ and strength S_σ in the low and high mass regimes. In the high mass regime the σ -meson is increasingly suppressed. Therefore either the strength of the interaction or the range has to increase, in order to still be able to bound the deuteron. On the other hand, in the low mass regime, the lightness of the meson lets the ψ_{np} and $\psi_{\sigma np}$ -part of the wave function become equally dominant¹⁰. A single meson would be easy to generate and annihilate, not contributing much to the binding energy, without scaling the range or strength parameters up.

It was found that a second meson leads to significant correction of the ground state energy. In the initial parameter setup, the second meson contributed with -0.173 MeV or 7% to the total binding energy, which is one magnitude smaller than the first meson's contribution, but may not be small enough to treat this perturbatively. A third meson would probably contribute with even one magnitude less to the total binding energy, which would 1% or less.

Varying the parameters like in the case for just one meson, the model show a similar behaviour, see Fig. 7. This supports the expectation that the second meson just yield a small energy correction. Since the σ -meson mass is relatively uncertain [3] and other exchange meson candidates exist [10], the mass varying second meson contribution, Fig. 8, is of especially interesting. The second meson's contribution to the ground state energy was found to increase for decreasing masses. This was expected to happen, since the cost in energy to generate a meson will fall.

The significance of pion-mass σ -mesons is more than double the contribution at a mass of 500 MeV. Even though the nucleon pion interaction is of a different kind, due to the spin and isospin character, it gives a clue about how important pions might be to the nuclear force. Due to the lower mass of the pions, the two pion contribution might be more important than the first σ -meson. The pion interaction would definitely be an extension to the model, worth to explore.

¹⁰The larger the mass terms in the diagonal of the matrix in Eq. (28), the smaller the significance to the eigenvalues.

The second meson correction to the wave function, Fig. 9, is a small. The wave function was found to be symmetric by the exchange of the mesons. The model seems to automatically produce symmetric wave function, due to a large parameter space, ensured by appropriate large number of Gaussians.

8 Conclusion

In [1] a nuclear model is proposed, where the nucleons find themselves in a superposition with mesons. The nuclear model was applied to the deuteron using up to 2 scalar-isoscalar σ -mesons. The deuteron is then in a superposition of a proton-neutron, a proton-neutron- σ and a proton-neutron- σ - σ state. The result of solving the two-, three- and four-body systems with the numerical correlated Gaussian method, is a bound deuteron, where varied parameters of range, b_σ , strength S_σ and meson mass m_σ lead to varying significance of the second meson to the ground state energy. Especially with decreasing masses, not in the very light regime, the significance rose. The deuteron's radial wave function obtained in the 2-meson model is the same as in the 1-meson model plus a small correction. The radial wave function was found to be symmetric by the exchange of the two mesons, so the model's results satisfy quantum mechanical conditions.

A next step could be the inclusion of a third σ -meson to obtain accuracy of about 0.01 MeV of the binding energy, since it is expected to contribute with less than 1% to the binding energy. Furthermore the computational load is not expected to inflate, since the eigenenergy already converges with about 5 Gaussians in the 2-meson subsystem.

Another outlook would be the inclusion of pions. Due to their lower mass and the result of low mass second σ -meson computations, they are expected to play an important role. Even the contribution of multiple pions could be of greater importance than the σ -meson's.

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